

Project Proposal

Automated Outreach, Meeting Management, and Student Engagement Tools

Bridge School

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1. Problem Introduction

Effective communication and task management are essential for the smooth operation of educational institutions. Bridge School currently relies on a shared Excel spreadsheet to record all meeting minutes and decisions. As the school grows, this approach has become increasingly restrictive: it is difficult to search through past decisions, track who is responsible for what, and ensure that follow-ups actually occur. Important action items are buried in spreadsheet rows, leading to a lack of accountability and lost information. In addition, the school's outreach and parent communication workflows remain largely manual, which is time-consuming and prone to missed deadlines.

The objective of this project is to build a focused, human-centric tech solution addressing five core needs: an AI-powered meeting workspace and action tracker, an email automation pipeline, professional email templates, a WhatsApp notification channel for parents, and a Student & Parent Portal. In addition, we will develop an Automated Performance Broadcaster to streamline parent-teacher communication on student progress.

2. Implementation & Features

2.1 AI-Powered Meeting Workspace & Action Tracker

A centralised web application to replace the current spreadsheet system for logging meeting notes and tracking decisions. Staff can record meeting notes, tag specific colleagues with action items, and monitor task progress. An AI integration will automatically generate a short executive summary from long meeting notes, reducing the administrative burden after each session. We will also conduct an onboarding session to introduce and assist non-technical staff in getting familiar with the application.

Key features include:

- **Centralised Digital Archive:** Meeting notes stored and indexed by date, department, or meeting type, making historical decisions easily searchable.
- **Smart Action Item Management:** Users can tag specific staff members with action items during a meeting. The system tracks the status of each task (e.g., *Pending, In Progress, Completed*).
- **Automated Reminders:** An integrated notification system (via email or a dashboard alert) that pings staff when an action item deadline is approaching or overdue.
- **AI-Powered Auto-Summarisation:** Integration with an AI API to automatically generate a short executive summary from long meeting notes.

Technical details: The application will be built with a React frontend and a FastAPI (Python) backend, with data stored in a MySQL database. Reminders will be delivered via the Gmail API. The AI summarisation feature will use Gemini 3.1 Flash-Lite, which provides a free daily request quota. The feature will remain available as long as the daily limit has not been reached. Should Bridge School’s usage exceed the free tier, the system can be configured to use a paid model if it is feasible.

2.2 Email Automation Pipeline

An automation tool that allows Bridge School to schedule emails in advance and trigger follow-up messages based on specific events or dates. The pipeline will also support AI-assisted content suggestions to help staff draft routine communications.

Technical details: Email scheduling and dispatch will be handled via the Gmail API. A task scheduler (Python’s APScheduler library) will manage scheduled sends. The AI-assisted drafting feature will use Gemini 3.1 Flash-Lite, which provides a free daily request quota. The feature will remain available as long as the daily limit has not been reached. Should Bridge School’s usage exceed the free tier, the system can be configured to use a paid model if it is feasible.

2.3 Visually Engaging Email Templates

A set of professionally designed, responsive HTML email templates for newsletters, announcements, and outreach campaigns. Templates will be easy to edit by non-technical staff through clearly labelled placeholder fields.

Technical details: Templates will be built in standard HTML/CSS compatible with major email clients (Gmail, Outlook). They will be integrated directly into the automation pipeline (Section 2.2) so staff can select a template, fill in content, and dispatch — all from one interface.

2.4 Automated Performance Broadcaster

A tool that allows teachers to upload student result sheets (CSV or Excel format), which the system then uses to automatically generate and send personalised performance reports to parents via email. This eliminates the manual effort currently involved in communicating student progress individually.

Technical details: A CSV/Excel parser (Python’s `pandas` library) will process uploaded result sheets. Personalised email bodies will be generated per student record and

dispatched through the same Gmail API pipeline built in Section 2.2, sharing the existing infrastructure with no significant additional overhead.

2.5 WhatsApp Notification Integration

A complementary notification channel that delivers key alerts to parents via WhatsApp, acknowledging that many parents in Pakistan use WhatsApp as their primary means of communication over email. Performance report notifications, action item reminders for staff, and notice board updates will all be deliverable through this channel in addition to email.

Technical details: Notifications will be sent using the WhatsApp Business API via Twilio, which provides a free sandbox environment suitable for Bridge School's scale. The integration will plug directly into the existing notification infrastructure built in Sections 2.1 and 2.4, requiring no significant additional backend work.

2.6 Student & Parent Portal

A dedicated, login-protected web portal giving students and parents direct access to key information without needing to contact the school. The portal will serve as a centralised hub for progress tracking, school communications, and opportunities.

Key features include:

- **Student Progress Dashboard:** A personalised view showing the student's academic performance, attendance record, and historical result reports, drawing from data already managed through Section 2.4.
- **Notice Board:** A digital board where school management can post announcements, events, and updates, visible to all portal users in real time.
- **Opportunities & Scholarships:** A curated, regularly updated section listing relevant national and international opportunities, competitions, and scholarships for students.

Technical details: The portal will be built as an extension of the existing React frontend, with role-based access (student, parent, staff) managed via JWT authentication through the FastAPI backend. Student data will be drawn from the same MySQL database used across the system, ensuring consistency with no duplication of data.

3. System Integration

All components will be designed to integrate with Bridge School's existing workflows with minimal friction:

- **Email:** The system will connect to an existing Gmail or Google Workspace account via the Gmail API. No migration of existing email data is required.
- **WhatsApp:** The WhatsApp Business API will be connected to an existing or newly registered WhatsApp Business number for Bridge School. The Twilio sandbox is sufficient at the school's current scale and involves no additional cost.

- **Student & Parent Portal:** The portal will be accessible through any modern web browser. Parents and students will receive their login credentials via email upon setup. No app installation is required.
- **Meeting Records:** The meeting workspace will replace the existing Excel spreadsheet system entirely. All historical records can be imported into the new system at the start, ensuring no data is lost during the transition.
- **Hosting:** The application backend and database will need to be hosted on a server. This can be an existing server already available at Bridge School, or a low-cost cloud server (e.g. a Virtual Private Server) which can be procured at a modest monthly cost. We will advise Bridge School on the most suitable option based on their existing setup.
- **Access:** The dashboard will be accessible through any modern web browser — no software installation is needed on staff devices.
- **Handover:** Full source code and a deployment guide will be delivered at the end of the project, ensuring ease of future maintenance.

4. Staff Training Plan

We recognise that the system’s value depends on staff being able to use it confidently. The following training steps are built into Phase 4:

- **User guide:** A concise written guide (with screenshots) covering all core workflows: logging meeting notes, assigning and tracking action items, scheduling emails, selecting templates, uploading result sheets, managing the notice board, and posting opportunities. We will also produce a thorough walkthrough video guiding staff through every feature of the system.
- **Onboarding session:** A live session with the relevant staff members demonstrating all features end-to-end, conducted in-person or online based on convenience.
- **Post-handover support:** We will remain available for a period of two weeks after handover to address any questions or minor issues via email.

5. Tentative Timeline

The project is planned to start on 1 June 2026 and will be completed by 20 July 2026. The 200 total hours (50 hours per member) are distributed across evenings and weekends.

Phase	Dates	Est. Hours	Deliverables
Phase 1	1 June – 5 June	~28 hrs	SRS written. API contracts and ERDs designed. UI designs produced. Tech stack confirmed and environments set up.
Discovery & Setup			
Phase 2	6 June – 28 Jun	~80 hrs	Meeting Workspace functional (archive, action items, reminders). Email templates coded. Gmail API integrated. Student & Parent Portal built (dashboard, notice board, opportunities). JWT authentication implemented.
Core Development			
Phase 3	29 Jun – 13 Jul	~68 hrs	Email automation pipeline complete (scheduling, triggers, AI drafting). AI summarisation integrated. Performance Broadcaster complete. WhatsApp notifications integrated.
Integration			
Phase 4	14 Jul – 20 Jul	~24 hrs	End-to-end testing and bug fixes. User guide and walkthrough video produced. Onboarding session conducted. Full handover to Bridge School.
Testing & Handover			
Total	1 June – 20 Jul	200 hrs	

We welcome any feedback on the scope or features before work begins, and are happy to adjust priorities to best suit Bridge School's needs.